

Milestone 1 Findings

Literature review · framework assessment · data compendium

A clean source-led deck: every finding uses an excerpt rendered directly from the cited paper, report, or official page.

Seven themes

What the primary literature converges on



LITERATURE REVIEW

SOURCE-LED FINDING

No existing framework was designed for TRL 1–3

ORIGINAL SOURCE EXCERPT · p. 26

II. EIC Pathfinder

The overall objective of the EIC Pathfinder for advanced research is to develop the scientific basis to underpin breakthrough technologies. It provides support for the earliest stages of scientific, technological or deep tech research and development. Pathfinder projects aim to build on new, cutting-edge directions in science and technology to disrupt a field and a market or create new opportunities by realising innovative technological solutions through:

- ✦ 'EIC Pathfinder Open', open to support projects in any field of science, technology or application without predefined thematic priorities.
- ✦ 'EIC Pathfinder Challenges' to support coherent portfolios of projects within predefined thematic areas with the aim to achieve specific objectives for each Challenge.

II.1 EIC Pathfinder Open

- ✦ Do you have an ambitious vision for a novel future technology that could make a real difference to our lives?
- ✦ Do you see a plausible way of achieving the scientific breakthrough that will make this technology possible?
- ✦ Can you imagine collaborating with an interdisciplinary team of researchers and

COMMENTS

- What the excerpt shows: the EU's own 'earliest stage' instrument, Pathfinder, opens at the earliest stages of deep tech research — the stage is officially recognised.
- But recognition is not a method: Pathfinder funds research consortia, not market discovery. The gap is procedural, not conceptual.
- Question for interviews: who owns market discovery at TRL 1–3 in Switzerland? Cf. the Innosuisse stage slide (24) and the CRL slides (18–20).

European Commission (2026), EIC Work Programme 2026, p. 26 (EIC Pathfinder).

LITERATURE REVIEW

SOURCE-LED FINDING

Ecosystem, not policy, drives deep tech

ORIGINAL SOURCE EXCERPT · p. 1

The Journal of Technology Transfer (2026) 51:2886–2915
<https://doi.org/10.1007/s10961-026-10327-w>

RESEARCH



Building deep-tech entrepreneurial ecosystems: factors that promote the presence of deep-tech startups in European regions

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Abstract
Deep-tech startups aim to implement and commercialize novel technologies that have the potential to address societal challenges. Due to current geopolitical trends and the potential of such technologies to rejuvenate the industrial power of regions, European governments are attempting to foster the creation of this type of startups within their borders. However, they struggle to do so effectively due to a limited understanding of the factors that support the emergence of deep-tech startups. We combine the generic entrepreneurial ecosystem framework with specific deep-tech enabling elements to develop and test hypotheses about which factors influence the presence of deep-tech startups on a regional scale. We develop a thesaurus to identify deep-tech startups at the regional level and test our hypotheses using quantitative analyses in 272 European NUTS-2 regions located in 27 EU countries and the UK. We show that the presence of specific deep-tech enabling knowledge and talent is more important for facilitating the emergence of deep-tech startups than the quality of the generic entrepreneurial ecosystem and the presence of specific deep-tech enabling

COMMENTS

- What the excerpt shows: the study's own abstract — specific knowledge and talent matter more for deep-tech startup emergence than formal institutions.
- Deep tech clusters where research infrastructure and specialised people already are; policy quality alone does not move it.
- Muoniverse reading: the PSI/ETH/EPFL density is the asset to build on. Open question: does that talent pool cover commercial roles, not only physics? Cf. slide 31.

Micol, F., Leendertse, J. & van Rijnsoever, F. (2026), Journal of Technology Transfer 51, pp. 2886–2915.

LITERATURE REVIEW

SOURCE-LED FINDING

TTOs: relationships beat transactions

ORIGINAL SOURCE EXCERPT · p. 16

The evolving role of technology transfer offices in the entrepreneurial...1075

ing risky, socially important basic research over commercially infeasible timeframes. TTOs support both goals through building relationships within the wider university innovation service ecosystem, operating in support of the university's social charter.

Further, supporting the evolutionary economics perspective (e.g., Freeman and Soete, 1997) the findings show that TNCs create selection pressure; shaping upstream innovation through their buying power, reinforcing Goel and Göktepe-Hultén (2018)'s observation that *"Due to the dominance of big established firms in getting access to academic knowledge, the nature of the research may be skewed and lead to narrower patents."* (p.254). In this respect, the findings highlight the challenges presented to university research through short-termism in corporate governance and capital markets, and the retreat of private finance from R&D (Mazzucato & Semieniuk, 2017; Pisano, 2006). Therefore, debate about how to fund innovation is important – particularly innovation that addresses grand challenges affecting environment (5H1) and civil society (5H2) such as clean energy, quality of life, food security, and climate change, which require transformative (i.e., disruptive and risky) approaches unlikely to be funded from private sources.

6 Conclusion

Addressing the calls to investigate technology transfer within the university innovation service ecosystem more holistically (de Falani Bezerra & Torkomian, 2023; Good et al., 2019; Miller et al., 2018), this study emphasises the importance of a wider view of quintuple helix (5H) stakeholders including society and the environment. This study extends current knowledge about how TTOs facilitate stakeholder relationships through the technology commercialization process, by integrating university competencies and market demand. The research highlighted that TTOs' power is increasing, as research follows funding. TTOs' economic power is growing through relational ties, and through access to 'home-grown' financial resources. Thus, TTOs are rapidly becoming playmakers in a dynamic and rapidly evolving university innovation service ecosystem.

As universities are configured for exploration rather than exploitation, public-private partnership is the logical solution to realising non-economic (i.e., environmental and social) returns from research, particularly in the life sciences. While taxpayers and governments carry the risk of frontier-extending research, care is required to ensure that returns reflect

COMMENTS

- What the excerpt shows: Chen et al.'s term 'playmakers' — TTOs that translate science through relationships across the innovation ecosystem.
- Transfer behaves as an orchestration capability, not a licensing-desk function; the survival panel reaches the same conclusion independently (slide 34).
- Interview probe: is PSI's knowledge-transfer office resourced and measured for orchestration, or for transactions (patents, licences)?

Chen, Z., Little, V. & Thuan, N. (2025), Journal of Technology Transfer 50, pp. 1060–1079.

LITERATURE REVIEW

SOURCE-LED FINDING

Structured frameworks work — if adapted

ORIGINAL SOURCE EXCERPT · p. 1

economic development, prompting extensive research into support programs—such as venture funds, accelerators, and proof-of-concept centers—designed to support academic entrepreneurship. Yet, many university scientists lack the entrepreneurial skills, experience, and networks required to navigate commercialization pathways effectively. In response, a new generation of government-supported programs has emerged globally, with a primary focus on early-stage entrepreneurial development. This study examines the origins, design, and evolution of such initiatives across eight countries, most of which are modeled on the U.S. National Science Foundation’s Innovation Corps (I-Corps) program but have been adapted to reflect regional contexts and institutional needs. Through cross-case analysis, we demonstrate that these programs constitute a multi-tiered infrastructure that strengthens the translation of academic research into market-ready innovations. These initiatives are increasingly vital to the advancement of entrepreneurial universities and the maturation of national and regional innovation ecosystems. The paper concludes with comparative insights and suggestions for future research. For policymakers and program designers, the findings highlight how program choices can be used as levers to strengthen entrepreneurial capacity, align interventions with the diverse needs of scientists, and improve the integration of customer discovery and related forms of training withing broader ecosystems.

Keywords Academic entrepreneurship · Commercialization · University spinoffs · Lean LaunchPad · Lean startup method · Customer discovery · Entrepreneurship ecosystems

JEL Classification O31 · O33 · O57 · L26 · M53 · I23

COMMENTS

- What the excerpt shows: Hayter et al. trace I-Corps-style programs — structured customer discovery — spreading across national systems.
- A standardised methodology travels, and every adopter adapts it locally: structure wins over ad hoc, but not unchanged.
- Design tension for Milestone 2: methodology-driven (I-Corps) vs mentor-driven (Innosuisse, slide 22). Which fits a physics NCCR?

Hayter, C. S. et al. (2025), Journal of Technology Transfer, DOI 10.1007/s10961-025-10288-6.

LITERATURE REVIEW

SOURCE-LED FINDING

Configuring support matters more than providing it

ORIGINAL SOURCE EXCERPT · p. 32

Springer

Table 7 Subgroup analysis on program structure, service offer, and sponsorship

	k	n	Effect size	95% CI		weight	z	Prob≥z	sig	Q/QB	Prob≥Q	I ² (%)	τ ²
Overall	65	10,159	0.067	0.022	0.112	100.00%	2.89	0.0038	*	355.08	0.0000	85.20	0.0263
Program structure & service offer													
Type										5.65	0.059		
Business incubator	24	4869	−0.011	−0.102	0.080	37.14%	−0.24	0.8128	ns	195.56	0.0000	89.35	0.0434
University incubator	10	1520	0.118	0.029	0.205	14.6%	2.60	0.0092	**	19.62	0.0204	58.96	0.0108
Science Park	22	3770	0.110	0.058	0.162	32.34%	4.11	0.0000	***	42.84	0.0033	54.38	0.0074
Specialization										0.02	0.878		
Specific	40	5674	0.070	0.010	0.129	58.54%	2.29	0.0223	*	153.53	0.0000	78.74	0.0272
Generic	25	8989	0.063	−0.008	0.133	41.10%	1.74	0.0823	ns	200.19	0.0000	90.18	0.0266
Provision of services													
Network													
Yes	60	13,899	0.071	0.024	0.118	91.92%	2.93	0.0034	**	337.62	0.0000	85.72	0.0268
Financing										1.86	0.173		
Yes	34	9829	0.043	−0.031	0.115	54.22%	1.14	0.2537	ns	292.35	0.0000	91.67	0.0397
No	25	4025	0.102	0.057	0.147	36.63%	4.41	0.0000	***	40.05	0.0211	43.63	0.0051
R&D										1.34	0.247		
Yes	39	10,174	0.054	−0.012	0.120	60.65%	1.59	0.1112	ns	296.88	0.0000	90.07	0.0363
No	21	3725	0.103	0.052	0.155	31.27%	3.91	0.0001	***	39.51	0.0058	53.01	0.0067
Office space										1.64	0.200		
Yes	58	13,457	0.069	0.019	0.118	88.44%	2.73	0.0064	***	335.92	0.0000	86.20	0.0279
No	2	442	0.137	0.044	0.228	3.48%	2.88	0.0039	**	0.49	0.4843	0.00	0.0000
Mentoring										1.34	0.247		
Yes	39	10,174	0.054	−0.012	0.120	60.65%	1.59	0.1112	ns	296.88	0.0000	90.07	0.0363
No	21	3725	0.103	0.052	0.155	31.27%	3.91	0.0001	***	39.51	0.0058	53.01	0.0067
Age of incubator										1.95	0.378		
≥5 and <10	5	2190	0.099	0.041	0.156	9.4%	3.33	0.0009	***	7.69	0.1035	48.00	0.0021
≥10 and <20	7	683	0.115	0.039	0.189	9.03%	2.96	0.0031	**	6.60	0.3598	0.00	0.0000
≥20	10	702	0.183	0.078	0.284	12.96%	3.41	0.0007	***	17.71	0.0386	49.26	0.0145

N. Seitz et al.

Seitz, N. et al. (2026), Review of Managerial Science, DOI 10.1007/s11846-026-01017-w, Table of moderator effects.

COMMENTS

- What the excerpt shows: the moderator rows — university incubators ($r = 0.118$) outperform generic ones; mentoring's coefficient is slightly negative (-0.012).
- The variable is not support volume but configuration: who hosts it and what it connects you to. Negative mentoring reads as opportunity cost of founder time.
- Pairs with the advisor coefficients on slide 33 — two independent studies question 'add more mentors' as a strategy.

LITERATURE REVIEW

SOURCE-LED FINDING

Market is the binding constraint

ORIGINAL SOURCE EXCERPT · p. 10

market (EFC2: outer loading = 0.828) has the highest impact on enhanced foresight capabilities (EFC);

The structural model outlines that market features (MF) have the strongest impact enhanced foresight capabilities (EFC) (path coefficient = 0.514), followed by environmental features (EF) on firm's internal features (FIF) (path coefficient = 0.480). The lowest impact on enhanced foresight capabilities (EFC) corresponds to technological features (path coefficient = 0.046) – [Fig. 3](#).

The variables FIF, EF, MF and TF explain 47 % of the variance in the case of EFC (R-square = 0.470), EF and TF explain 35.6 % of the variance in the case of FIF (R-square = 0.356), while EF and TF explain 40.7 % of the variance in the case of MF (R-square = 0.407).

The square root of each construct's AVE is greater than its highest correlation with other constructs (observed under the main diagonal), demonstrating the discriminant validity of the latent variables based on the Fornell-Larcker criterion ([Table 4](#)).

COMMENTS

- What the excerpt shows: in the structural model, market features drive foresight capability hardest (path = 0.514), ahead of technology features.
- The mechanism is capability-building: market signals teach firms to see where the technology could go — firm-internal factors were not significant.
- For muon applications with no existing market, that foresight process must be organised deliberately. Cf. the flipped-Lean answer, slide 17.

Capatina, A., Bleoju, G. & Kalisz, D. (2024), Journal of Innovation & Knowledge 9, 100515.

LITERATURE REVIEW

SOURCE-LED FINDING

Commercial exposure from day one

ORIGINAL SOURCE EXCERPT · p. 2

1556

U. Cantner et al.

process into distinct phases, starting with the research phase and concluding with the establishment of the spin-off. Venturing scientists need to transition along these phases. We observe that the level of embeddedness in both spheres influences the success of these transitions. Furthermore, the commercial sphere holds greater importance than the academic sphere, right from the outset of the ASO process. To support the creation of spin-offs, policymakers should focus on facilitating scientists' exposure to the commercial sphere. This can be achieved by implementing entrepreneurship education initiatives and encouraging scientists to gain industry experience. Additionally, academic institutions can play a vital role in supporting scientists by reducing administrative burdens and recognizing their entrepreneurial efforts alongside their academic qualifications. Future research could expand our understanding of the relative importance of both spheres in other contexts, such as social entrepreneurship, where commercial and social-oriented logics converge.

Keywords Academic spin-off · Conflicting logics · Process perspective · Phase model · Dominance analysis

JEL Classification L26 · O31 · O33

1 Introduction

Academic spin-offs (ASOs) are an important mechanism shown that in this dynamic and multi-phase process, founders need to accomplish a specific set of activities in each development phase before progressing to the next (Ndonzuau et al., 2002, Rasmussen, 2011, van Geenhuizen & Soetanto, 2009, Vohora et al., 2004, Wood, 2011). Academic entrepreneurs must overcome “critical junctures”, defined as complex problems that “occur at a point along a new high-tech venture’s expansion path, preventing it from achieving the transition from one development phase to the next” (Vohora et al., 2004, p. 159). These critical junctures arise because different phases of spin-off development require distinct configurations of resources, capabilities, network ties, and support. Qualitative studies document how ASOs develop through the creation process depending on the academic entrepreneurs’ access to specific resources and social networks in different process phases (Fernández-Alles et al., 2015, Hayter, 2016a, b). Additionally, first quantitative analyses focus on specific phases of the ASO creation process. For instance, Krabel & Mueller (2009) explored the drivers of individual academic scientists’ decision to pursue ASO creation, i.e., becoming nascent entrepreneurs, while Landry et al. (2006) investigated the individual and organizational assets that increase the likelihood of ASO formation. However, a quantitative assessment of the entire ASO creation process, its different phases, and the transitions between these phases is still missing. Such an analysis would provide insights into the relevant determinants in each phase and can have implications

COMMENTS

- What the excerpt shows: in Cantner's panel, the commercial sphere outweighs the academic sphere from the very outset of spin-off creation.
- This falsifies 'research first, commercialise later' sequencing — commercial capability is built in parallel, not afterwards.
- Hard question for muons: what is commercial exposure when the product is 10–20 years out? A Milestone 2 interview anchor; cf. slide 41.

Cantner, U. et al. (2024), Small Business Economics 62, pp. 1555–1590.

LITERATURE REVIEW

SOURCE-LED FINDING

Advisor type matters more than quantity

ORIGINAL SOURCE EXCERPT · p. 19

Help, I need somebody! Business and technology advisors in...

1585

lowers AR by 0.39 ($p=0.05$), which means that for every (unstandardized) additional technology advisor, AR goes down by about 0.31. We thus find support for the negative relation between technology advisors and AR hypothesized in Hypothesis 1b. The magnitude of these effects suggests that roughly 4 business advisors can advance AR by a full unit, while only 3 technology advisors are needed to delay AR by a full unit in the same amount of time. Notably, the direct effects of experience align with those of advisors in sign, but their effect is much smaller: about 15 years of additional entrepreneurial experience would be needed to advance AR with 1, and about 20 years of technology experience is needed to delay AR with 1.³ Advisors generate a much more substantial effect on AR than SBV founders' own experience.

Hypothesis 2 argued that founder experience in a similar knowledge domain as the advisor would negatively moderate the effect of advice on AR. Model 3 (Table 3) shows a significant, negative coefficient of -0.18 ($p=0.04$) for the technology advisor-technology experience combination. The negative effect of technology advisors is especially strong for entrepreneurs high in technology experience (Fig. 2a). These results of the technology advice-technology experience combination are aligned with H2. On the other hand, we do not see the hypothesized negative effect of the business advisor-entrepreneurial experience combination, as it has a significant, positive coefficient of 0.38 ($p<0.005$) in Model 3 (Table 3 and Fig. 2b): the effect of business advisors is more positive for founders high in

COMMENTS

- What the excerpt shows: each business advisor raises application readiness by ~ 0.23 ; each technology advisor lowers it by ~ 0.31 .
- Counterintuitive but consistent with Theme 4: at early stages the scarce input is business framing, not more technical validation.
- Implication for the NCCR advisor pool: recruit advisors who have priced deep-tech markets, not more professors. Cf. slide 7.

Denoo, L., Van Boxstael, A. & Belz, A. (2024), Journal of Technology Transfer 49, pp. 1567–1605.

LITERATURE REVIEW

SOURCE-LED FINDING

Europe's structural barriers

ORIGINAL SOURCE EXCERPT · p. 3

Despite the region's leadership in research - with seven of the top ten spots in the Global Innovation Index and nine of the world's top 20 research institutions - they continue to fall short in commercialization and investment.

Share of VC Funding in Key Deep Tech areas, 2021-2023

Technology Area	Europe	US	China	Rest of world
Energy Technologies	28%	32%	15%	25%
Quantum Technologies	30%	35%	10%	25%
Robotics & Autonomous Systems	35%	40%	15%	10%
Advanced Materials, Manufacturing & Recycling Technologies	38%	42%	10%	10%
Space & Propulsion Technologies	30%	35%	10%	25%
Advanced Sensing Technologies	30%	35%	15%	20%
Biotechnologies (Genetics & Synbio)	30%	35%	15%	20%
AI Technologies	30%	35%	15%	20%
Advanced Connectivity, Navigation and Digital Technologies	30%	35%	15%	20%
Advanced Semiconductor Technologies	30%	35%	15%	20%

Source: Cambridge, Switzerland, Venture Catalyst (2023)

Over the past five years, European Deep Tech firms secured just €58 billion in funding, compared to €215 billion in the U.S. In fact, less than 10 per cent of DeepTech unicorns are based in Europe.

Europe's long-term competitiveness is at stake here. That's why in this article, with the help from [Linn Kretzschmar](#), CERN Venture Connect Programme Lead at CERN, [Ann-Lauriene Schumacher](#), Entrepreneurship Manager at ETH AI Center, and [Manuel Hösle](#), Coach & Investment Manager at [Fraunhofer Venture](#), we will explain why the region is falling behind in Deep Tech commercialization and, more importantly, what should be done about it.

Challenges and gaps in Deep Tech commercialization | Stryber

COMMENTS

- What the excerpt shows: European deep tech firms secured a fraction of US capital — €58B vs €215B over the report's window.
- The gap is structural — investor risk appetite and failure stigma, not just fund volume; capital alone cannot close it.
- Switzerland leads per capita (slide 35), but late-stage rounds still face EU-scale limits; cf. restored EIC access, slide 26.

Stryber (2025), Deep Tech commercialization in Europe: the gap, the challenge, and the fix.

LITERATURE REVIEW

SOURCE-LED FINDING

Deep tech needs 35% more time, 48% more capital

ORIGINAL SOURCE EXCERPT · p. 54

...increasing time-to-market associated with deep tech startups secured the third of all startup funding on the continent in 2023⁹³, underscoring the sector’s increasing importance in driving technological development and economic growth.

Despite growing investor interest, deep tech startups continue to face funding barriers. Studies indicate that deep tech startups often require 35% more time and 48% more capital than traditional startups to achieve comparable revenue levels of \$5 million or more⁹⁴. Insights from stakeholder interviews and findings from the regional snapshot section highlight limited investment access as a persistent barrier. Unlike other startups that may achieve rapid scaling with relatively low capital, deep tech ventures demand substantial upfront investment in R&D, prototyping, testing, and regulatory approvals before reaching market viability. These long development cycles mean that deep tech startups often take years to commercialize, requiring investors who are willing to adopt a patient and long-term approach.

A challenge is the high level of risk and uncertainty associated with deep tech investments. Many deep tech innovations are scientifically complex, unproven at scale, and require significant technical validation before they can be deployed in real-world applications. This high-risk profile—combined with long return timelines—makes deep tech less attractive to traditional VC firms that typically seek faster exits and lower-risk investments. The disconnection between scientific

COMMENTS

- What the excerpt shows: deep tech takes on average 35% more time and 48% more capital to reach equivalent milestones.
- That premium is an average across all deep tech — physics ventures sit further into the tail of both distributions.
- Design constraint for every instrument in this deck: BRIDGE runs 12 months, VC funds 3–5 years. Cf. slides 37 and 41.

UNDP (2025), Global Deep Tech Ecosystems: Catalyzing Innovation for Sustainable Development, p. 54.

LITERATURE REVIEW

SOURCE-LED FINDING

Converging on what deep tech is

ORIGINAL SOURCE EXCERPT · p. 1

Deep-Tech Innovation: A Multi-Method Study toward a Conceptual Framework and Research Agenda

Johann Kortsch¹, Stefan Raff-Heinen^{2,3}, David Bendig¹, Martin Murmann², Colin Schulz¹, Fiona Murray³

Abstract

The term “deep-tech innovation” has attracted growing attention in research and practice, but it is applied inconsistently and lacks an agreed-upon definition. This limits cumulative knowledge building and blurs how deep-tech innovation relates to adjacent concepts. We address this gap by developing a framework that treats deep-tech innovation as a distinct object of inquiry. Using a multi-method design that combines a systematic, integrative, concept-centric literature review and semi-structured interviews with deep-tech founders, we identify twelve defining attributes structured across three levels: invention, venture, and ecosystem. At the invention level (the conceptual core), we specify six attributes: three foundational attributes that capture the scientific and technological basis of the invention (scientific origins, hardware intensity, and an enabling platform character), and three attributes that describe its characteristic exposure profile (long development timelines, high capital intensity, and multilayered

COMMENTS

- What the excerpt shows: twelve defining attributes structured across technology, venture and ecosystem levels — the field is converging on a definition.
- Convergence is what makes cross-study comparison legitimate; muon technology satisfies the technology-side attributes by construction.
- Use the three levels as the interview sampling frame: each level names different actors to talk to.

Kortsch, J. et al. (2024), Deep-Tech Innovation: A Multi-Method Study, MIT Sloan / University of Münster.

Four frameworks, one gap

RCA / Imperial · CRL · Innosuisse · EIC · candidates



FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

The official pipeline starts at TRL 1 — but scale-up begins later

ORIGINAL SOURCE EXCERPT · p. 1



EIC Accelerator

What for	Who can apply	What you get
Develop and scaleup innovations Develop and scaleup innovations with the potential to create or disrupt markets (TRL 6-8). Patient capital principle.	Startups and SMEs Single start-ups and SMEs, individuals intending to launch a SME and small mid-caps (can apply for equity only).	Grants and investments Grant component below EUR 2.5 million. Up to € 10 million of equity investments, coaching & mentoring, networking.

What is the EIC Accelerator ?

The **EIC Accelerator** is a funding programme under Horizon Europe that offers support to start-ups and SMEs that:

- have a **innovative, game changing product**, service or business model that could **create new markets** or **disrupt existing ones** in Europe and even worldwide,
- have the ambition and commitment to **scale up**,
- are looking for **substantial funding**, but the risks involved are too high for private investors alone to invest

COMMENTS

- What the excerpt shows: the EU's flagship commercial instrument, the Accelerator, requires TRL 6–8 — demonstration, not discovery.
- The official pipeline funds research early (Pathfinder) but starts commercial support only at TRL 5–6: the commercial gap is precisely TRL 1–3.
- This slide is the thesis's central finding; the rest of the deck maps what exists on each side of that gap.

European Innovation Council (2026), EIC Accelerator official programme page.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

RCA/Imperial: a cluster, not a program

ORIGINAL SOURCE EXCERPT · p. 1

[Entrepreneurship](#)
[Research](#)
[Programmes and Engagement](#)
[DT Prime](#)
[Our People](#)

Turning breakthrough research into investable ventures

Combining the patience of a university alongside the commercial discipline of an investor, we created DT Prime specifically for Imperial College London. We work with researchers to turn deep tech and life science innovations into ventures that can scale, aligning the people, data, and strategy needed to attract top-tier capital.

DT Prime is particularly suited for staff-led ventures that require time, technical validation, and strategic support to reach commercial viability. Built around the specific demands of life science and deep tech, DT Prime management works with Imperial's researchers to provide a combination of non-dilutive funding and tailored support from experienced venture builders.

Every engagement is bespoke, with non-dilutive funding of £50,000–£150,000 in each phase deployed toward clearly defined value inflection points - from generating technical data required by investors and shaping an IP position, to validating market demand or preparing for investor conversations. High-performing teams return for multiple rounds, with cumulative support of up to £500,000.

Science and technology focus areas

We support ventures tackling complex, high-value problems where deep science can unlock new markets, reshape industries, and deliver lasting commercial and societal value.

Our focus is on technology domains with strong scientific foundations, meaningful market

COMMENTS

- What the excerpt shows: DT Prime's official page — turning breakthrough research into investable ventures with non-dilutive funding.
- It is a venture builder attached to a university, not a course: governance and capital sit alongside the methodology.
- Benchmark question for November: could an equivalent attach to PSI? The methodology transfers; the institution does not automatically.

Imperial College London, DT Prime, Institute for Deep Tech Entrepreneurship (official page).

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

RCA/Imperial: 'flipping' Lean Startup

ORIGINAL SOURCE EXCERPT · p. 8

[Raising Money](#)

Lean Meets Wicked Problems

Posted on July 25, 2023 by steve blank

This post [previously appeared](#) in Poets & Quants.

I just spent a month and a half at Imperial College London co-teaching a “Wicked” Entrepreneurship class. In this case Wicked doesn’t mean morally evil, but refers to really complex problems, ones with multiple moving parts, where the solution isn’t obvious. (Understanding and solving homelessness, disinformation, climate change mitigation or an insurgency are examples of wicked problems. [Companies also face Wicked problems](#). In contrast, designing AI-driven enterprise software or building dating apps are comparatively simple problems.)

I’ve known [Professor Cristobal Garcia](#) since 2010 when he hosted my first visit to Catholic University in Santiago of Chile and to southern Patagonia. Now at Imperial College Business School and Co-Founder of the [Wicked Acceleration Labs](#), Cristobal and I wondered if we could combine the tenets of Lean (get out of the building, build MVPs, run experiments, move with speed and urgency) with the expanded toolset developed by researchers who work on Wicked problems and Systems’ Thinking.

Our goal was to see if we could get students to [stop admiring problems and work rapidly on solving them](#). As Wicked and Lean seem to be mutually exclusive, this was a pretty audacious undertaking.

This five-week class was going to be our MVP.

COMMENTS

- What the excerpt shows: Blank's own account of adapting Lean for wicked problems — technology first, then the search for applications.
- The flip inverts classic lean: instead of starting from a customer problem, you start from a capability and hunt for contexts where it matters.
- Key open issue: what replaces customer discovery when no customer exists yet — Blank's answer is lead and extreme users.

Blank, S. (2023), Lean Meets Wicked Problems, Steve Blank blog / Imperial College Business School class account.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

CRL: the 9-point commercial scale

ORIGINAL SOURCE EXCERPT · p. 31

5. Proposed Model for the joint assessment of Technical and Commercial Readiness and Market Maturity

5.2 Commercialisation Readiness Levels (CRL) Overview and Rationale

The CRL will assess various indicators which influence the commercial and market conditions beyond the technology maturity. This enables key barriers to be addressed to support the commercialisation of a technology. It is intended to supplement the TRL scale by providing increased focus on a technology's preparedness for market launch and commercial success through specific and clearly defined business indicators.

Like the TRLs, the CRLs are on a scale from 1 – 9 to identify the commercial readiness of the technology, from basic value proposition stage (CRL-1) to trusted solution (CRL-9). The CRL approach was designed to give a framework for assessing a technology's commercial readiness, primarily for the goal of bringing it to market. This encompasses market analysis, cost-effectiveness, scalability, and regulatory compliance, which are often not addressed by the TRL system.

Commercialisation Readiness Level (CRL)	Requisite Conditions
CRL 1	Basic value proposition of technology identified and reported
CRL 2	Business concept formulated with potential applications
CRL 3	Business Plans validated with Proof-of-Business-Case
CRL 4	Minimum Viable Product Completed and Pilots initiated
CRL 5	Minimum Marketable Product and Operational Processes Validated
CRL 6	Minimum Marketable Product Deployed in the Market with Operating Revenue Targets
CRL 7	Matured Product Design and Marketing Strategy Validated
CRL 8	Matured Product Deployed in the Market with Targets Achieved
CRL 9	A Trusted Solution and a successful business model

COMMENTS

- What the excerpt shows: CRL codified 1–9, from basic value proposition to real market competitiveness — a government found it useful enough to operationalise.
- Practitioner traction despite no peer-reviewed origin is itself a finding: there is real demand for a commercial axis to complement TRL.
- CRL 1–3 (opportunity recognition → market sizing) maps onto the thesis's target stage; read against the TRL table, slide 39.

NITI Aayog (2023), A New Lens for Innovation in New India: TCRM Matrix, p. 31.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

CRL: the documentation problem

ORIGINAL SOURCE EXCERPT · p. 8

assumption that a proven prototype translates naturally into reliable production.

The Commercial Readiness Level framework addressed the third dimension. Developed by Professor Ali Abbas and Dr. Mobin Nomvar at the University of Sydney, and designed to operate in synchronisation with TRL, CRL asks whether the external world – customers, regulators, financiers, and the broader value chain – is prepared to accept and integrate a new technology. For those building businesses, these are critical questions.

Companies in Advanced Materials, for example,

Deep Tech Leaders (2025), TRL, MRL & CRL Explained, Part 1, p. 8.

COMMENTS

- What the excerpt shows: secondary sources describing and propagating CRL — none of them cites a primary, peer-reviewed source.
- Adoption by citation chain without a canonical paper: building analysis on it requires explicit methodological honesty.
- Thesis position: CRL is treated as a practitioner framework, not a validated instrument — stated openly in the assessment.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

CRL: the commercial dimension TRL misses

ORIGINAL SOURCE EXCERPT · p. 32

MRL 8	Product Market Fit Validated with Proven Scalability and Market Share Target Fixed
MRL 9	Continuous Iteration of Product Development through Feedback and Market Research

5.4 Identifying the gaps

Each of the Technology readiness level (TRL), Commercialization readiness level (CRL), and Market readiness level (MRL) systems, have certain limitations which are:

Technology Readiness Level (TRL)	Commercialization Readiness Level (CRL)	Market Readiness Level (MRL)
TRLs only take into account a technology's technical readiness, ignoring other elements like market demand, cost-effectiveness, and regulatory compliance that are crucial for commercial success. TRLs do not specify how to advance a technology from one level to the next, making it difficult to design and implement technological development and commercialization strategies.	CRLs are concerned with commercialization readiness and may overlook technical elements of a technology. External variables, such as changes in rules or market conditions, might have an impact on a technology's commercial readiness.	The MRL system may be subjective, and various stakeholders may perceive it differently, resulting in contradictions in ratings. MRLs may fail to account for external factors that can influence technology adoption, such as changes in customer tastes, competition, or technological improvements.

24

A New Lens for Innovation in New India -
Introducing the Techno-Commercial Readiness and Market Maturity Matrix

COMMENTS

- What the excerpt shows: even NITI's motivation mirrors the thesis — TRL accounts only for technical readiness, not commercial readiness.
- The criticism is validated; the remedy is thin: CRL describes where you are, not what to do next — no rubric or checklist exists.
- Milestone 2 opening: ask actors how they actually gauge commercial readiness — a detector can be TRL 9 and CRL 1.

NITI Aayog (2023), TCRM Matrix, p. 32 (identifying the limits of TRL and the need for commercial readiness).

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

Innosuisse: the native Swiss framework

ORIGINAL SOURCE EXCERPT · p. 1



Schweizerische Eidgenossenschaft
Confédération suisse
Confederazione Svizzera
Confederaziun svizra

Innosuisse
Swiss Innovation Agency

Start-up Coaching

The Start-up Coaching is aimed at young companies and their founders. It offers three different programmes, each focusing on specific development aspects in a way that is tailored to each start-up and comes with a voucher for the corresponding amount and validity period.
Coaching for young enterprises

INITIAL COACHING
Worth up to CHF 10,000

Challenging and optimising the business model

>

CORE COACHING*
Worth up to CHF 50,000

Creating and developing business traction

>

SCALE-UP COACHING
Worth up to CHF 75,000

Refining and implementing the growth strategy

* Condition: completed Initial Coaching

COMMENTS

- What the excerpt shows: the official Start-up Coaching offer aimed at young companies — the instrument every Swiss venture touches.
- The design is a voucher system: flexibility and founder choice, rather than a standardised curriculum.
- Cf. slide 28: Switzerland chose mentor-driven over methodology-driven — a deliberate contrast for the benchmarking.

Innosuisse (official), Start-up Coaching overview.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

Innosuisse: bespoke, voucher-based coaching

ORIGINAL SOURCE EXCERPT · p. 2

Innosuisse understands the importance of a diverse and inclusive ecosystem and works tirelessly to strengthen and develop the Swiss start-up landscape.

By joining the Innosuisse Start-up Coaching, you can benefit from reliable advice and potential referrals from investors and business partners by experienced entrepreneurs and experts.

Support offer Start-up Coaching

Initial Coaching

Initial Coaching grants you a voucher worth up to CHF 10 000 for coaching sessions with Innosuisse accredited coaches over a period of up to twelve months.

Core Coaching

Core Coaching grants you a voucher worth of up to CHF 50 000 for coaching sessions with Innosuisse accredited coaches over a maximum period of 36 months.

Scale-up Coaching

Scale-up Coaching grants you a voucher worth up to CHF 75 000 for coaching sessions with Innosuisse accredited coaches over a period of up to 24 months.

Contact

COMMENTS

- What the excerpt shows: initial coaching is a voucher (up to CHF 10,000) with which founders choose accredited coaches themselves.
- Demand-side design presupposes informed demand — at TRL 1–3, scientists may not yet know what support they need.
- Interview probe: how many accredited coaches have priced particle-physics applications? Supply of relevant expertise is the test.

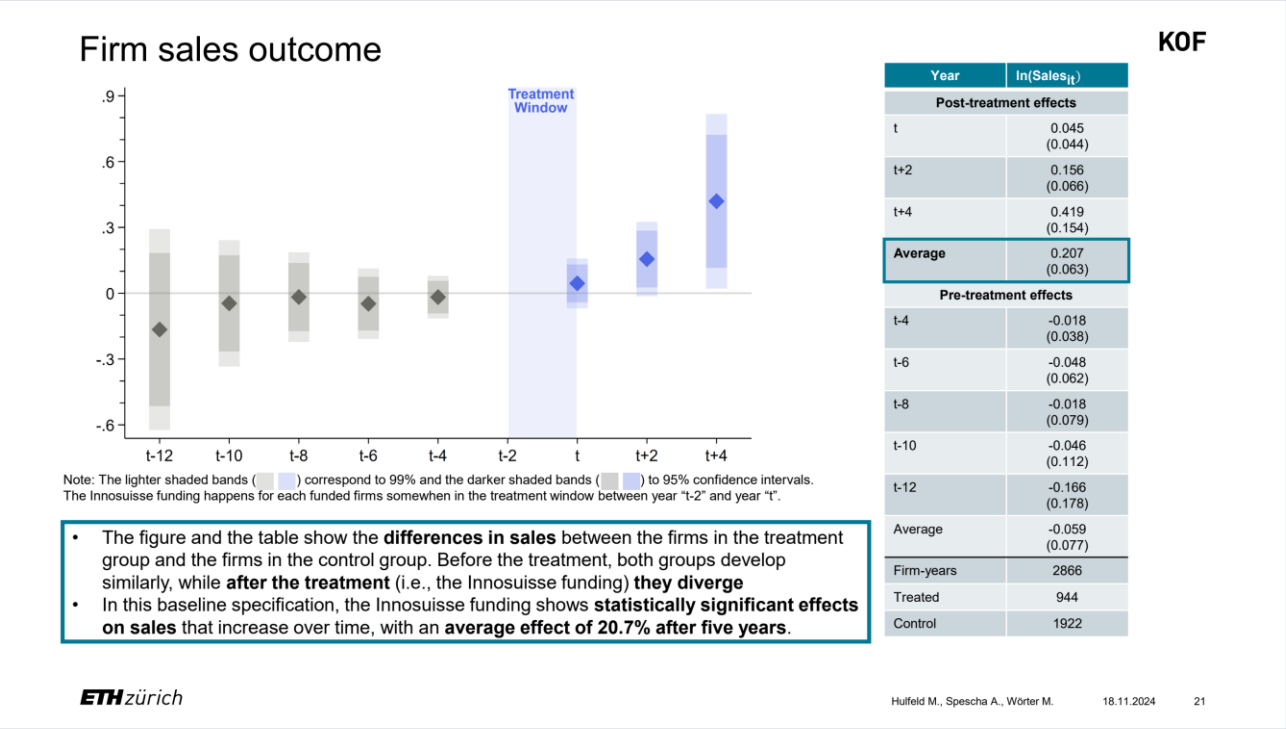
Innosuisse (official), Start-up Coaching overview, voucher-based programme description.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

Innosuisse: KOF-verified causal effect

ORIGINAL SOURCE EXCERPT · p. 21



COMMENTS

- What the excerpt shows: the KOF difference-in-differences estimate — +20.7% sales on average five years after funding.
- Rare causal evidence in this field, and the effect grows with time — consistent with patient, compounding returns.
- Boundary condition: strongest for small firms working with ETH-domain partners — evidence about firms, silent on pre-founding TRL 1–3.

Hulfeld, F., Spescha, A. & Wörter, M. (KOF ETH Zurich, 2024), Funding R&D Cooperation between Firms and Universities.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

Innosuisse support is stage-specific

ORIGINAL SOURCE EXCERPT · p. 2

Innosuisse understands the importance of a diverse and inclusive ecosystem and works tirelessly to strengthen and develop the Swiss start-up landscape.

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Scale-up Coaching

Scale-up Coaching grants you a voucher worth up to CHF 75 000 for coaching sessions with Innosuisse accredited coaches over a period of up to 24 months.

Contact

COMMENTS

- What the excerpt shows: the coaching ladder — initial, core, scale-up — where every rung presupposes an existing young company.
- Before founding there is no rung: the gap is stage mismatch inside a well-run system, not a missing system.
- With BRIDGE (slide 37): proof-of-concept money exists but runs 12 months — cf. UNDP's +35% time premium, slide 41.

Innosuisse (official), Start-up Coaching overview — stage-specific support and durations.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

EIC Accelerator: TRL 6–8

ORIGINAL SOURCE EXCERPT · p. 1



EIC Accelerator

What for Develop and scaleup innovations Develop and scaleup innovations with the potential to create or disrupt markets (TRL 6-8). Patient capital principle.	Who can apply Startups and SMEs Single start-ups and SMEs, individuals intending to launch a SME and small mid-caps (can apply for equity only).	What you get Grants and investments Grant component below EUR 2.5 million. Up to € 10 million of equity investments, coaching & mentoring, networking.
---	---	---

What is the EIC Accelerator ?

The **EIC Accelerator** is a funding programme under Horizon Europe that offers support to start-ups and SMEs that:

- have a **innovative, game changing product**, service or business model that could **create new markets** or **disrupt existing ones** in Europe and even worldwide,
- have the ambition and commitment to **scale up**,
- are looking for **substantial funding**, but the risks involved are too high for private investors alone to invest

COMMENTS

- What the excerpt shows: the Accelerator's own terms — TRL 6–8, blended finance (grant + equity), single-digit success rate.
- For muon ventures this is a later-stage target, not an entry point; the pathway runs Pathfinder → Transition → Accelerator.
- Sequencing, not any single instrument, is the operative design question; cf. Swiss eligibility, slide 26.

European Innovation Council (official), EIC Accelerator programme page.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

EIC: Switzerland fully eligible since Jan 2025

ORIGINAL SOURCE EXCERPT · p. 2

Europe programme on similar terms with EU Member States. They can now join and/or lead research consortia and be funded directly from the programme for all calls from the 2025 budget onwards, enabling them to contribute to cutting-edge research and tackle global challenges as well as engage in initiatives supporting national research reforms and institutional capacity-building.

[Frequently asked questions](#)

[Horizon Europe key figures for Switzerland](#)

[Country page](#)

Projects and results

Research project database (CORDIS)

The Commission's primary portal for results of EU-funded research projects. Here you can find projects related to Switzerland.

[EU funded projects](#)

Contact

National Contact Points

The [National Contact Points \(NCPs\)](#) provide guidance, practical information and assistance on all aspects of participation in Horizon Europe.

European Commission

[DG Research and Innovation unit 03: International cooperation](#)

Documents

30 DECEMBER 2014

COMMENTS

- What the excerpt shows: Switzerland associated to Horizon Europe 'on similar terms with EU Member States' — full eligibility restored.
- The EU pipeline is a real option again for Swiss ventures; recent EIC winners confirm actual uptake.
- Interview probe: which PSI-adjacent teams already coordinate EU consortia — capability, not just eligibility, gates access.

European Commission (official), International cooperation with Switzerland in research and innovation.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

Candidate: CERN Venture Connect

ORIGINAL SOURCE EXCERPT · p. 3

Despite the region's leadership in research - with seven of the top ten spots in the Global Innovation Index and nine of the world's top 20 research institutions - they continue to fall short in commercialization and investment.

Share of VC Funding in Key Deep Tech areas, 2021-2023

Technology Area	Europe	US	China	Rest of world
Energy Technologies	28%	32%	18%	22%
Quantum Technologies	18%	35%	12%	35%
Robotics & Autonomous Systems	35%	32%	25%	8%
Advanced Materials, Manufacturing & Recycling Technologies	38%	32%	15%	15%
Space & Propulsion Technologies	32%	35%	12%	19%
Advanced Sensing Technologies	35%	32%	15%	18%
Biotechnologies (Genetics & Synbio)	32%	35%	15%	18%
AI Technologies	25%	45%	35%	15%
Advanced Connectivity, Navigation and Digital Technologies	32%	35%	15%	18%
Advanced Semiconductor Technologies	28%	32%	15%	25%

Source: Cambridge, Stephenson, Wootton Catalyst (2023)

Over the past five years, European Deep Tech firms secured just €58 billion in funding, compared to €215 billion in the U.S. In fact, less than 10 per cent of DeepTech unicorns are based in Europe.

Europe's long-term competitiveness is at stake here. That's why in this article, with the help from [Linn Kretzschmar](#), CERN Venture Connect Programme Lead at CERN, [Ann-Lauriene Schumacher](#), Entrepreneurship Manager at ETH AI Center, and [Manuel Hösle](#), Coach & Investment Manager at Fraunhofer Venture, we will explain why the region is falling behind in Deep Tech commercialization and, more importantly, what should be done about it.

Challenges and gaps in Deep Tech commercialization | Stryber

COMMENTS

- What the excerpt shows: CERN's Venture Connect lead among the study's expert sources — CERN is actively building a venture path for particle physics.
- The closest institutional sibling to Muoniverse: same science, same pre-market problem, answered with a curated, zero-equity model.
- November benchmarking target: what transfers from CERN's portfolio curation to a Swiss NCCR context?

Stryber (2025), Deep Tech commercialization in Europe — source identifying CERN Venture Connect as a relevant institutional actor.

FRAMEWORK ASSESSMENT

SOURCE-LED FINDING

Candidate: NSF I-Corps

ORIGINAL SOURCE EXCERPT · p. 4

C. S. Hayter et al.

et al., 2017; Swamidass, 2013). Closely modeled on the Lean LaunchPad methodology developed by Silicon Valley entrepreneur Steve Blank, I-Corps aims to help academic researchers evaluate the commercial potential of their innovations through structured customer discovery and business model development (Blank & Dorf, 2012).¹ The program integrates Blank’s customer discovery process with Osterwalder’s BMC (Osterwalder & Pigneur, 2010), providing a systematic approach to identifying target markets and refining value propositions. Perhaps more importantly, NSF officials gravitated toward the program because they saw that it closely mirrored the scientific method and would therefore be familiar to academic entrepreneurs.²

Although few systematic evaluations have been conducted, descriptive and observational studies suggest that I-Corps enhances understanding among academic entrepreneurs of market segments dynamics, fosters a more entrepreneurial mindset, and accelerates the translation of university-based research into commercial outcomes (Arkilic, 2019; Huang-Saad et al., 2017; NSF, 2023). The program’s perceived success has prompted the development of similar initiatives in countries around the world (Arkilic, 2019). As discussed below, we demonstrate how I-Corps—and particularly its international adaptations—has played a central role in shaping how national and regional policymakers approach the support of academic entrepreneurship.

COMMENTS

- What the excerpt shows: I-Corps' core as Hayter et al. document it — structured customer discovery and business model development.
- It is a methodology rather than an institution: adoptable without new bodies, which makes it the cheapest candidate to pilot.
- Open question: does a 7-week customer-discovery sprint fit muon validation cycles? Cf. flipped Lean, slide 17.

Hayter, C. S. et al. (2025), Journal of Technology Transfer, p. 4.

The numbers

Statistical evidence behind the findings



SOURCE-LED FINDING

Market features dominate ($\beta = 0.514$)

ORIGINAL SOURCE EXCERPT · p. 10

market (EFC2: outer loading = 0.828) has the highest impact on enhanced foresight capabilities (EFC);

The structural model outlines that market features (MF) have the strongest impact enhanced foresight capabilities (EFC) (path coefficient = 0.514), followed by environmental features (EF) on firm's internal features (FIF) (path coefficient = 0.480). The lowest impact on enhanced foresight capabilities (EFC) corresponds to technological features (path coefficient = 0.046) – Fig. 3.

The variables FIF, EF, MF and TF explain 47 % of the variance in the case of EFC (R-square = 0.470), EF and TF explain 35.6 % of the variance in the case of FIF (R-square = 0.356), while EF and TF explain 40.7 % of the variance in the case of MF (R-square = 0.407).

The square root of each construct's AVE is greater than its highest correlation with other constructs (observed under the main diagonal), demonstrating the discriminant validity of the latent variables based on the Fornell-Larcker criterion (Table 4).

COMMENTS

- What the excerpt shows: the coefficient hierarchy — market features 0.514 vs technology ~0.31; firm-internal features not significant.
- Market-side variance explains the foresight capability outcome ($R^2 = 0.47$); the internal factors managers control directly did not register.
- Caveat for the thesis: cross-sectional PLS-SEM, $n = 110$ — association, not causation. Would it replicate for physics ventures?

Capatina, A., Bleoju, G. & Kalisz, D. (2024), Journal of Innovation & Knowledge 9, 100515, structural model.

DATA COMPENDIUM

SOURCE-LED FINDING

Talent and knowledge drive deep tech

ORIGINAL SOURCE EXCERPT · p. 40

Building deep-tech entrepreneurial ecosystems: factors that promote the...2911

Dependent variable: presence of deep-tech startups

	(1)	(2)	(3)	(4)	(5)	(6)
Specific knowledge			1.35 ***	1.21– 1.52		1.45 ***
Specific talent				1.32 **	1.11– 1.56	1.26 **
Specific formal institutions					0.97 1.02	0.92 0.87– 0.98
Observations	272	272	272	272	272	272
McFadden pseudo R ²	0.231	0.233	0.250	0.240	0.234	0.262
AIC	1288.746	1287.417	1262.449	1278.397	1288.203	1245.026

* p<0.05

** p<0.01

*** p<0.001

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We would like to acknowledge the great help provided by Fabia Beckers from Utrecht University for important reasoning carried forward in the first stages of the paper (especially regarding deep-tech definition and for the help in building the thesauruses). We would like to acknowledge also the great help provided by Arnaud de la Tour from Hello Tomorrow for giving us access to their taxonomy, Lech Bakhuizen van den Brink and Arjan Goudblom from Techleap for their early feedback and suggestions for complementing our thesauruses, and to Crunchbase for providing access to the data. We appreciated also the comments made by the anonymous reviewers, which helped us further clarify some concepts and overall improve the paper.

Author contributions

Federico Micol: Conceptualization, Data curation, Methodology, Software, Formal analysis, Visualization, Writing—original draft, Writing—review and editing.Jip Leendertse: Conceptualization, Data curation, Methodology, Software, Formal analysis, Visualization, Writing—original draft, Writing—review and editing. Frank van Rijnsoever: Supervision, Conceptualization, Methodology, Writing—review and editing.

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Data availability

The Dataset is provided based on the request.

Declarations

COMMENTS

- What the excerpt shows: the regression table — specific knowledge and talent significant; formal institutions not, across 272 regions.
- The significant predictors are slow variables: research infrastructure and people accumulate over decades, not funding cycles.
- Swiss reading: one very strong node (PSI/ETH/EPFL) — is that concentration a leverage point or a single-point fragility?

Micol, F., Leendertse, J. & van Rijnsoever, F. (2026), Journal of Technology Transfer, Table 4.

DATA COMPENDIUM

SOURCE-LED FINDING

Incubation: type and age matter

ORIGINAL SOURCE EXCERPT · p. 32

Springer

Table 7 Subgroup analysis on program structure, service offer, and sponsorship

	k	n	Effect size	95% CI		weight	z	Prob≥z	sig	Q/QB	Prob≥Q	I ² (%)	τ ²
Overall	65	10,159	0.067	0.022	0.112	100.00%	2.89	0.0038	*	355.08	0.0000	85.20	0.0263
Program structure & service offer													
Type										5.65	0.059		
Business incubator	24	4869	−0.011	−0.102	0.080	37.14%	−0.24	0.8128	ns	195.56	0.0000	89.35	0.0434
University incubator	10	1520	0.118	0.029	0.205	14.6%	2.60	0.0092	**	19.62	0.0204	58.96	0.0108
Science Park	22	3770	0.110	0.058	0.162	32.34%	4.11	0.0000	***	42.84	0.0033	54.38	0.0074
Specialization										0.02	0.878		
Specific	40	5674	0.070	0.010	0.129	58.54%	2.29	0.0223	*	153.53	0.0000	78.74	0.0272
Generic	25	8989	0.063	−0.008	0.133	41.10%	1.74	0.0823	ns	200.19	0.0000	90.18	0.0266
Provision of services													
Network													
Yes	60	13,899	0.071	0.024	0.118	91.92%	2.93	0.0034	**	337.62	0.0000	85.72	0.0268
Financing										1.86	0.173		
Yes	34	9829	0.043	−0.031	0.115	54.22%	1.14	0.2537	ns	292.35	0.0000	91.67	0.0397
No	25	4025	0.102	0.057	0.147	36.63%	4.41	0.0000	***	40.05	0.0211	43.63	0.0051
R&D										1.34	0.247		
Yes	39	10,174	0.054	−0.012	0.120	60.65%	1.59	0.1112	ns	296.88	0.0000	90.07	0.0363
No	21	3725	0.103	0.052	0.155	31.27%	3.91	0.0001	***	39.51	0.0058	53.01	0.0067
Office space										1.64	0.200		
Yes	58	13,457	0.069	0.019	0.118	88.44%	2.73	0.0064	***	335.92	0.0000	86.20	0.0279
No	2	442	0.137	0.044	0.228	3.48%	2.88	0.0039	**	0.49	0.4843	0.00	0.0000
Mentoring										1.34	0.247		
Yes	39	10,174	0.054	−0.012	0.120	60.65%	1.59	0.1112	ns	296.88	0.0000	90.07	0.0363
No	21	3725	0.103	0.052	0.155	31.27%	3.91	0.0001	***	39.51	0.0058	53.01	0.0067
Age of incubator										1.95	0.378		
≥5 and <10	5	2190	0.099	0.041	0.156	9.4%	3.33	0.0009	***	7.69	0.1035	48.00	0.0021
≥10 and <20	7	683	0.115	0.039	0.189	9.03%	2.96	0.0031	**	6.60	0.3598	0.00	0.0000
≥20	10	702	0.183	0.078	0.284	12.96%	3.41	0.0007	***	17.71	0.0386	49.26	0.0145

N. Seitz et al.

COMMENTS

- What the excerpt shows: the moderator rows — university incubators 0.118 vs generic 0.067; incubators older than 20 years reach 0.183.
- Networks compound: incubator age nearly doubles the effect — evidence the real asset is accumulated relationships.
- A new PSI-hosted incubator starts at age zero; renting maturity through partners (CERN, ETH) may beat building it. Cf. slide 27.

Seitz, N. et al. (2026), Review of Managerial Science, moderator table.

SOURCE-LED FINDING

Advisors: business +0.23, technology -0.31

ORIGINAL SOURCE EXCERPT · p. 19

Help, I need somebody! Business and technology advisors in...

1585

lowers AR by 0.39 ($p=0.05$), which means that for every (unstandardized) additional technology advisor, AR goes down by about 0.31. We thus find support for the negative relation between technology advisors and AR hypothesized in Hypothesis 1b. The magnitude of these effects suggests that roughly 4 business advisors can advance AR by a full unit, while only 3 technology advisors are needed to delay AR by a full unit in the same amount of time. Notably, the direct effects of experience align with those of advisors in sign, but their effect is much smaller: about 15 years of additional entrepreneurial experience would be needed to advance AR with 1, and about 20 years of technology experience is needed to delay AR with 1.³ Advisors generate a much more substantial effect on AR than SBV founders' own experience.

Hypothesis 2 argued that founder experience in a similar knowledge domain as the advisor would negatively moderate the effect of advice on AR. Model 3 (Table 3) shows a significant, negative coefficient of -0.18 ($p=0.04$) for the technology advisor-technology experience combination. The negative effect of technology advisors is especially strong for entrepreneurs high in technology experience (Fig. 2a). These results of the technology advice-technology experience combination are aligned with H2. On the other hand, we do not see the hypothesized negative effect of the business advisor-entrepreneurial experience combination, as it has a significant, positive coefficient of 0.38 ($p<0.005$) in Model 3 (Table 3 and Fig. 2b): the effect of business advisors is more positive for founders high in entrepreneurial experience. Overall, we find mixed support for H2.

Hypothesis 3 suggested that knowledge variety in the advisor-experience combinations

COMMENTS

- What the excerpt shows: the substitution rate — roughly four business advisors advance application readiness by one full unit.
- Advisor design becomes arithmetic: the question is not whether advisors help, but which type and in what ratio.
- Boundary condition: $n = 112$ ventures; the outcome is readiness, not survival. What a business advisor does pre-product is untested.

Denoo, L., Van Boxstael, A. & Belz, A. (2024), Journal of Technology Transfer, p. 1585.

DATA COMPENDIUM

SOURCE-LED FINDING

TTOs become playmakers through relationships

ORIGINAL SOURCE EXCERPT · p. 16

The evolving role of technology transfer offices in the entrepreneurial...1075

ing risky, socially important basic research over commercially infeasible timeframes. TTOs support both goals through building relationships within the wider university innovation service ecosystem, operating in support of the university's social charter.

Further, supporting the evolutionary economics perspective (e.g., Freeman and Soete, 1997) the findings show that TNCs create selection pressure; shaping upstream innovation through their buying power, reinforcing Goel and Göktepe-Hultén (2018)'s observation that *"Due to the dominance of big established firms in getting access to academic knowledge, the nature of the research may be skewed and lead to narrower patents."* (p.254). In this respect, the findings highlight the challenges presented to university research through short-termism in corporate governance and capital markets, and the retreat of private finance from R&D (Mazzucato & Semieniuk, 2017; Pisano, 2006). Therefore, debate about how to fund innovation is important – particularly innovation that addresses grand challenges affecting environment (5H1) and civil society (5H2) such as clean energy, quality of life, food security, and climate change, which require transformative (i.e., disruptive and risky) approaches unlikely to be funded from private sources.

6 Conclusion

Addressing the calls to investigate technology transfer within the university innovation service ecosystem more holistically (de Falani Bezerra & Torkomian, 2023; Good et al., 2019; Miller et al., 2018), this study emphasises the importance of a wider view of quintuple helix (5H) stakeholders including society and the environment. This study extends current knowledge about how TTOs facilitate stakeholder relationships through the technology commercialization process, by integrating university competencies and market demand. The research highlighted that TTOs' power is increasing, as research follows funding. TTOs' economic power is growing through relational ties, and through access to 'home-grown' financial resources. Thus, TTOs are rapidly becoming playmakers in a dynamic and rapidly evolving university innovation service ecosystem.

As universities are configured for exploration rather than exploitation, public-private partnership is the logical solution to realising non-economic (i.e., environmental and social) returns from research, particularly in the life sciences. While taxpayers and governments carry the risk of frontier-extending research, care is required to ensure that returns reflect

COMMENTS

- What the excerpt shows: the playmaker mechanism — relationships within the wider university innovation-service ecosystem.
- Two literatures converge: Chen supplies the mechanism; the survival panel (business relationships, $p = 0.0034$) supplies the magnitude.
- Measurement implication: a TTO's KPIs could count ties built and introductions made — not only patents filed.

Chen, Z., Little, V. & Thuan, N. (2025), Journal of Technology Transfer, p. 1075.

SOURCE-LED FINDING

Switzerland: #1 per capita, 3× the runner-up

ORIGINAL SOURCE EXCERPT · p. 3

annual flow of capital into Swiss Deep Tech multiplied to 1.9 billion USD, with 2.3 billion projected for 2025. For 60 percent of all domestic venture funding, the terms Switzerland ranks first in Europe and third worldwide.

structural realignment. A decade ago, science-based startups formed a narrower startup scene. Today they anchor the capital market. The shift also influences public discourse: when it comes to 'deep tech', statistical reality now dictates that they are often seen as the future.

Share of total VC funding allocated in Deep Tech

COMMENTS

- What the excerpt shows: 60% of Swiss venture funding flows to deep tech — the highest share of any country.
- The ecosystem is deep-tech-native: capital preference, per-capita spinout value (100 vs 33 for the runner-up), top-ranked universities.
- Tension with slides 40–41: strength at formation — but will Swiss capital hold for 10–20-year physics timelines? An interview anchor.

Deep Tech Nation Switzerland / Dealroom.co (2025), Swiss Deep Tech Report 2025, via Swiss Startup Association reproduction.

DATA COMPENDIUM

SOURCE-LED FINDING

Innosuisse Innovation Projects: 57% launch rate

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Innovation projects with implementation partners

ational innovation projects are Innosuisse's most important funding organisations from the economy and society cooperate with research innovative products, services, processes or new business models.

- **Market launch:** Three years after the funding has ended, more innovations (57%) have either been launched on the market, or are scheduled to be launched in the next two years.
- **Job creation:** A funded project leads to an average of 2.5 new jobs over the years 2021–2023. Three years after the funding completion, the project will have created around 800 to 900 full-time positions. This has also been confirmed by a scientific study conducted by the Institute at ETH Zurich. Three years after completion, companies receiving Innosuisse funding report 18% higher employment than companies without Innosuisse funding.
- **The leverage effect of funding creates added value:** Through funding leads directly to additional added value at the funding completion. Funding brings the economy directly around four francs of added value per franc of funding and totalled for three further years.
- **Strengthening of knowledge and technology transfer:** The transfer of knowledge (KTT) also has a long-term effect: almost half of the

COMMENTS

- What the excerpt shows: 57% of innovation projects reach market launch within three years of funding ending.
- Solid conversion — but the sample is existing firm–university R&D consortia; selection happens before the statistic.
- The metric counts launches, not scale — and for muon ventures the whole 3-year window is pre-market. Cf. slide 41.

Innosuisse (2025), Impact monitoring 2021–2023.

SOURCE-LED FINDING

BRIDGE PoC: 70% founding rate

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Innosuisse
Swiss Innovation Agency

Innovation cheque

The innovation cheque offers SMEs and start-ups low-threshold funding. The aim is to carry out feasibility and market studies as well as technological clarifications.

- **Cooperation with research:** Thanks to the innovation cheque, more than half of the participating companies can be introduced to knowledge and technology transfer with research. 70% of SMEs continue to work with their research partner after the project is completed.
- **Results achieved:** The collaboration with research partners is successfully tested in almost all cases. 63% of companies have successfully tested and further developed their innovation idea and fully achieved this goal. Around a quarter of innovation cheques subsequently carry out an innovation project.

→ Detailed results and charts

Projects by researchers

Innovation projects without implementation partner

Innovation projects without implementation partners aim to develop scientific findings into applications for high-risk projects with high innovation potential and to take the first steps towards implementation.

- **Application orientation:** Half of the funded projects provide conclusive results for further implementation steps and half of the projects are highly likely to have longer-term potential for a market launch.
- **Collaboration with implementation partners:** More than two-thirds of the projects have already worked with potential implementation partners. Start-ups and spin-offs that emerge from research play an important role in KTT. In 38% of the projects, such a spin-off has already been or is planned.

→ Detailed results and charts

BRIDGE Proof of Concept

The support offers from the Swiss National Science Foundation (SNSF) and Innosuisse provide researchers with a year of support to develop research results into marketable products and prepare a career outside of research and universities.

- **Founding start-ups from research:** Around 70% of the supported researchers found a start-up or spin-off after the funding (start-up with formal IP licensing).

→ Detailed results and charts

Project set-up assistance and networking

Innovation Booster

COMMENTS

- What the excerpt shows: around 70% of BRIDGE-supported researchers go on to found a startup — the highest conversion of any instrument.
- Willingness is not the bottleneck: when an instrument fits the stage, researchers do found companies.
- The 12-month, CHF 130K format proves the concept — and its boundary: muon validation cycles outlast the window.

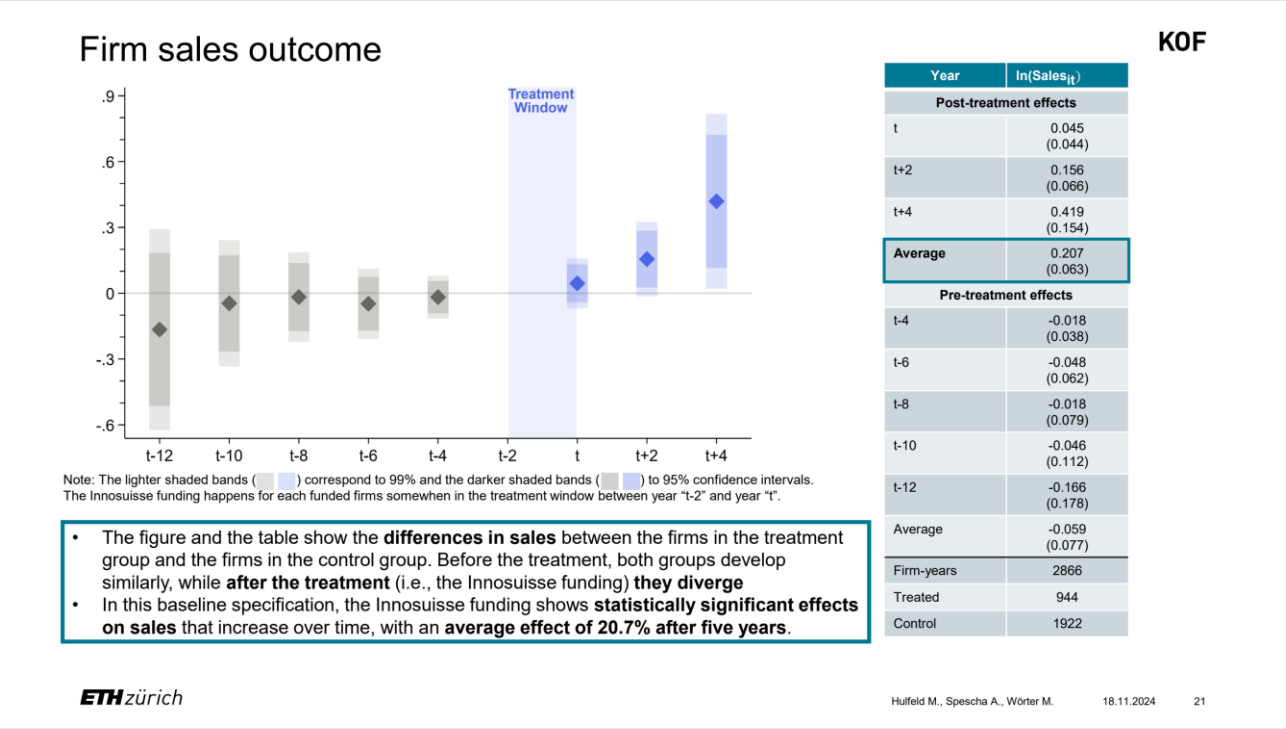
Innosuisse (2025), Impact monitoring 2021–2023, BRIDGE Proof of Concept.

DATA COMPENDIUM

SOURCE-LED FINDING

KOF causal effect: +20.7% sales, +17.6% employment

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COMMENTS

- What the excerpt shows: the 20.7% average sales effect; within the panel it rises from +7.8% (yr 0) toward ~28% (yr 4).
- Compounding returns argue for patient evaluation: a short measurement window would score these projects as failures.
- Same boundary as slide 23: significant for small firms with ETH-domain partners; silent on the pre-founding stage this thesis targets.

Hulfeld, F., Spescha, A. & Wörter, M. (KOF ETH Zurich, 2024), causal difference-in-differences result.

SOURCE-LED FINDING

The TRL 1–3 gap, visualized

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5. Proposed Model for the joint assessment of Technical and Commercial Readiness and Market Maturity

5.1 Technology Readiness Levels (TRL) Overview and Rationale

TRL is a framework for estimating the technology maturity of core technologies in a program during the selection process and in subsequent monitoring and evaluation phases until these technologies, or products utilizing them, attain market readiness. Originally introduced by NASA (National Aeronautics and Space Administration), the TRL scale is a metric with nine technology readiness levels for describing the maturity of a technology from ideation stage (TRL-1) to highest degree of application in its operational environment (TRL-9).

It is based on the assumption that as a technology advances from basic concept to final implementation, its maturity level increases. The scale is intended to give a standard language for innovators, researchers, and investors to communicate about the readiness of a technology for commercialization or deployment.

Table 2: Technology Readiness Levels (TRL) Scale

Technology Readiness Level (TRL [®])	Requisite Conditions
TRL 1	Basic principles observed and reported
TRL 2	Technology concept and/or application formulated
TRL 3	Analytical and Experimental Critical Function and/or Characteristic Proof-of-Concept
TRL 4	Technology Component / sub-system validation in laboratory environment
TRL 5	Technology Component / sub-system validation in relevant environment (industrially relevant environment in case of key enabling technologies)
TRL 6	Technology sub-system or prototype demonstration in a relevant environment
TRL 7	Technology System Prototype demonstration in an operational

COMMENTS

- What the excerpt shows: the TRL scale itself — from 'basic principles observed' to 'proven in operations'.
- Nothing in the nine levels asks whether anyone would buy the technology: it measures maturity of proof, not of market.
- Read side-by-side with the CRL table (slide 18): together they give the thesis its two-axis readiness framing.

NITI Aayog (2023), TCRM Matrix, p. 30 (TRL scale).

DATA COMPENDIUM

SOURCE-LED FINDING

EU vs US: a 3.7× capital gap

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Despite the region's leadership in research - with seven of the top ten spots in the Global Innovation Index and nine of the world's top 20 research institutions - they continue to fall short in commercialization and investment.

Share of VC Funding in Key Deep Tech areas, 2021-2023

Technology Area	Europe	US	China	Rest of world
Energy Technologies	28%	32%	15%	25%
Quantum Technologies	34%	31%	2%	33%
Robotics & Autonomous Systems	39%	31%	2%	28%
Advanced Materials, Manufacturing & Recycling Technologies	38%	23%	2%	37%
Space & Propulsion Technologies	30%	61%	2%	7%
Advanced Sensing Technologies	30%	30%	15%	25%
Biotechnologies (Genetics & Synbio)	30%	33%	15%	22%
AI Technologies	26%	43%	15%	16%
Advanced Connectivity, Navigation and Digital Technologies	24%	40%	15%	21%
Advanced Semiconductor Technologies	1%	20%	51%	28%

Source: Cambridge, Switzerland, Venture Catalyst (2023)

Over the past five years, European Deep Tech firms secured just €58 billion in funding, compared to €215 billion in the U.S. In fact, less than 10 per cent of DeepTech unicorns are based in Europe.

Europe's long-term competitiveness is at stake here. That's why in this article, with the help from [Linn Kretzschmar](#), CERN Venture Connect Programme Lead at CERN, [Ann-Lauriene Schumacher](#), Entrepreneurship Manager at ETH AI Center, and [Manuel Hösle](#), Coach & Investment Manager at [Fraunhofer Venture](#), we will explain why the region is falling behind in Deep Tech commercialization and, more importantly, what should be done about it.

Challenges and gaps in Deep Tech commercialization | Stryber

COMMENTS

- What the excerpt shows: European deep tech secured €58B against the US €215B — with investor behaviour explaining part of the difference.
- Stacked on the +48% capital premium (slide 41), the distance to the same milestone is multiplicative, not additive.
- Switzerland's 60% deep-tech share (slide 35) is the national answer — yet Series B scale still leaks to larger pools. Interview probe.

Stryber (2025), Deep Tech commercialization in Europe.

DATA COMPENDIUM

SOURCE-LED FINDING

Deep tech: +35% time, +48% capital

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...increasing time and capital requirements for European deep tech startups secured the amount of startup funding on the continent in 2023⁹³, underscoring the sector’s increasing importance in driving technological development and economic growth.

Despite growing investor interest, deep tech startups continue to face funding barriers. Studies indicate that deep tech startups often require 35% more time and 48% more capital than traditional startups to achieve comparable revenue levels of \$5 million or more⁹⁴. Insights from stakeholder interviews and findings from the regional snapshot section highlight limited investment access as a persistent barrier. Unlike other startups that may achieve rapid scaling with relatively low capital, deep tech ventures demand substantial upfront investment in R&D, prototyping, testing, and regulatory approvals before reaching market viability. These long development cycles mean that deep tech startups often take years to commercialize, requiring investors who are willing to adopt a patient and long-term approach.

A challenge is the high level of risk and uncertainty associated with deep tech investments. Many deep tech innovations are scientifically complex, unproven at scale, and require significant technical validation before they can be deployed in real-world applications. This high-risk profile—combined with long return timelines—makes deep tech less attractive to traditional VC firms that typically seek faster exits and lower-risk investments. The disconnection between scientific

COMMENTS

- What the excerpt shows: the UNDP quantification — 35% more time and 48% more capital for equivalent milestones.
- Every instrument in this deck is shorter than the asset it supports: coaching 36 months, BRIDGE 12, VC funds 3–5 years.
- This number is the design constraint the thesis answers with sequencing — see the EIC pathway, slides 25–26.

UNDP (2025), Global Deep Tech Ecosystems, p. 54.

MILESTONE 1

Source-led evidence → interview questions

The practical conclusion: early-TRL physics commercialization is not missing ideas — it is missing a support architecture designed for the gap.

Next: Milestone 2 interviews · September 1, 2026

All source excerpts are embedded from the primary PDFs/pages listed beneath each image.